Melissa Torgbi

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A dedicated PhD researcher exploring generalisation in Large Language Models (LLMs). Leveraging 4 years of industry experience in Natural Language Processing and experience working with clients on Responsible AI in multiple sectors.

Research Interests: Natural Language Processing, Large Language Models, Mechanistic Interpretability, Reasoning, Generalisation, Explainability, Interpretability, Responsible AI.

RELEVANT RESEARCH EXPERIENCE

PhD Research | Evaluating and Improving Generalisation in Foundation Models **2023 - Present**

- Investigates the robustness of Large Language Models in novel scenarios, focusing on how a model's reliance on pre-training data impacts concepts like complex reasoning and social bias.
- Techniques currently being explored to evaluate and improve models include diagnostic datasets grounded in cognitive linguistics, targeted fine-tuning and mechanistic interpretability.
- Presented research at the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL) 2025.
- Skills Demonstrated: experimental design, interdisciplinary collaboration, fine-tuning, academic report writing.

Lead Research Assistant |University of Bath & Wyser Ltd **2023-2024**

- Rapidly upskilled in Speech Processing to collaborate with Wyser's R&D team on mitigating linguistic biases in Automatic Speech Recognition (ASR) systems.
- Led research framing and experimental design, fine-tuning Whisper to improve speech recognition accuracy for regional UK dialects and vulnerable populations.
- First-authored two peer-reviewed papers published within a six-month timeframe and presented findings in the VarDial workshop at COLING 2025.
- Skills Demonstrated: Research framing, experimental design, automatic speech recognition, fine-tuning, research dissemination.

RELEVANT WORK EXPERIENCE

Senior Associate Data Science | SAS Software Limited **2019 - 2023**

- Data Scientist working across multiple sectors including banking, insurance, retail, manufacturing, and government.
- UK lead for Natural Language Processing in the Data Science team with experience using Linguistic Rules, Topic Modelling, Machine Learning, Word Embeddings, Deep Learning and Transformer-based Models.
- Using skills Data Analytics and Machine Learning to deliver value to clients and a UK representative for Responsible AI focusing on model explainability and model monitoring.
- Delivered guest lectures and workshops at Universities covering Data Visualisation, Machine Learning, Responsible AI and Data Ethics.
- Extensive experience liaising with stakeholders, understanding problems and communicating complex concepts to different audiences.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C++, C, MATLAB, JavaScript, Base SAS, SAS Viya

Python Libraries and Frameworks: Pandas, NumPy, NLTK, Gensim, OpenCV, Scikit-Learn, Keras, PyTorch

TEACHING AND SUPERVISORY EXPERIENCE

Student Project Supervision | University of Bath **2025 - Present**

Mentored MSc and BSc students through Natural Language Processing research projects, guiding research direction, methodology, and academic paper writing.

Computer Science and AI Lecturer | Pathway to Bath, University of Bath **2023 - Present**

Designed and delivered Computer Science and AI programmes for Year 12 students as part of a widening participation initiative.

Guest Lecturer| University of Bath **2024-Present**

Delivered guest lectures and unit content lectures for the Undergraduate Natural Language Processing module as well as a guest lecture for visiting prospective students in AI working with Student Recruitment & Admissions.

Workshop Facilitator| MEDAL Summer School, University of Birmingham **2025**

Created content for and facilitated an interactive coding workshop in Python: Probing Large Language Models for Linguistic and Constructional Information.

Tutor| University of Bath **2024-2026**

Lab demonstrator students at undergraduate and masters level in Natural Language Processing and Analytic Software Technologies.

EDUCATION

PhD Accountable, Responsible and Transparent AI, Computer Science | University of Bath **2023 - Present**

Key Modules: Natural Language Processing, Reinforcement Learning, Machine Learning

Project Title: Evaluating and Improving Generalisation in Foundation Models

MSc Digital and Technology Solutions Specialist: Data Analytics (Distinction) | Aston University **2020 - 2023**

Key Modules: Data Analysis

Individual Project Title: Non-invasive Monitoring of Beehives using Weight, Temperature and Humidity Sensors

BEng Electronic Engineering (1st) | University of Southampton **2016 - 2019**

Key Modules: Foundations of Machine Learning, Signal and Image Processing, Advanced Computer Architecture, Mathematics for Electrical and Electronic Engineering, Advanced Proramming

Individual Project Title: Convolutional Neural Network for the Detection of Apples

VOLUNTEERING

Speaker | Pint of Science **2026**

Presented an AI talk to a general public audience titled "Can your AI chatbot think outside the box?"

Coding Mentor | V Code **2026**

Mentored secondary school girls at a one-day coding workshop to promote women in tech.

Volunteer | Datakind UK **2022 - Present**

Using data skills to help charities get value from their data as a Data Ethics Advisor, Data Ambassador and DataDive Volunteer.

Podcast Production Team Member | Women in Data **2022 - Present**

Working in a team of volunteers to edit podcast episodes to a production ready level.

AWARDS

University of Bath Doctoral Recognition Award | Student Winner 2026
Black Tech Acheivemnet Award | Rising Star Finalist 2022

PUBLICATIONS

Scivetti, W.*, Torgbi, M.*, Shichman, M., Pellegrin, T., Blodgett, A., Bonial, C. and Madabushi, H.T., 2025, December. Beyond Memorization: Assessing Semantic Generalization in Large Language Models Using Phrasal Constructions. In Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (pp. 1184-1201).

Jones, M.D., Torgbi, M. and Tayyar Madabushi, H., 2026. Improving the Understandability of Clinical Guidelines: Development and Evaluation of a GPT-4–Based Pipeline. Journal of Medical Internet Research, 28, p.e81915.

Torgbi, M., Clayman, A., Speight, J.J. and Madabushi, H.T., 2025, January. Adapting Whisper for regional dialects: Enhancing public services for vulnerable populations in the United Kingdom. In Proceedings of the 12th Workshop on NLP for Similar Languages, Varieties and Dialects (pp. 29-38).

Imperial, J.M., Barayan, A., Stodden, R., Wilkens, R., Sánchez, R.M., Gao, L., Torgbi, M., Knight, D., Forey, G., Jablonkai, R.R. and Kochmar, E., 2025, November. Universalcefr: Enabling open multilingual research on language proficiency assessment. In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (pp. 9714-9766).

Bonial, C., Pellegrin, T., Torgbi, M. and Madabushi, H.T., 2025, September. From form to function: A constructional nli benchmark. In Proceedings of the Second International Workshop on Construction Grammars and NLP (pp. 172-179).

Wheeler, S., Torgbi, M. and Madabushi, H.T., 2025, September. Reducing Environmental Costs whilst Maintaining Operational Effectiveness in Large Language Models through Chain Routing. In Proceedings of the 21st Conference on Natural Language Processing (KONVENS 2025): Workshops (pp. 250-273).

Blodgett, A., Bonial, C., Pellegrin, T.A., Torgbi, M. and Madabushi, H.T., 2025, July. Illuminating Logical Fallacies with the CAMPFIRE Corpus. In Proceedings of the 19th Linguistic Annotation Workshop (LAW-XIX-2025) (pp. 285-296).

Torgbi, M., Clayman, A., Speight, J., Hirst, J. and Madabushi, H.T., 2025, August. Inclusive ASR for Critical Public Services: Debiasing with Actor-Simulated Speech. In International Conference on Text, Speech, and Dialogue (pp. 331-342). Cham: Springer Nature Switzerland.

Bonial, C., Torgbi, M., Weissweiler, L., Blodgett, A., Beuls, K., Van Eecke, P. and Madabushi, H.T., 2025, September. Proceedings of the Second International Workshop on Construction Grammars and NLP. In Proceedings of the Second International Workshop on Construction Grammars and NLP.

Smith, L., Bocheva, D., Brown, O., Lowery, C., Tarpy, H., Torgbi, M., Worajitwannakul, S. and Madabushi, H., Large Language Models May Promote Radicalization via Exposure to Extremist Content and Politicized Communities.